

Reinforcement Learning — Extended Learning Activity 1

Units II & III: MDPs, DP & Monte Carlo Methods | M.Tech AI & DS | VNR VJIET

Name: _____ Roll No.: _____

Q1. A self-driving car completes trips (pickup $\rightarrow$ dropoff, done). A thermostat controls room temperature 24/7 without stopping. Which task is *episodic* and which is *continuing*? For the continuing task, explain why we must use a discount factor $\gamma < 1$ when computing returns.

Q2. An agent receives rewards $R_1 = 1$, $R_2 = 2$, $R_3 = 3$ over a 3-step episode with $\gamma = 0.5$. (a) Compute G_2 . (b) Use the recursive relation $G_t = R_{t+1} + \gamma G_{t+1}$ to compute G_1 and G_0 .

Q3. An agent has computed $V_\pi(s)$ for all states but does not know the environment's transition probabilities. Can it use V_π to directly select actions? What function does it need instead, and why?

Q4. A 2-state MDP has state A always transitioning to terminal state B with reward $+4$. Start with $V(A) = V(B) = 0$ and $\gamma = 1$. Compute $V(A)$ after one Bellman backup of policy evaluation.

Q5. In policy iteration, the policy is updated greedily after evaluation converges. (a) Under what condition does policy iteration stop? (b) A classmate proposes improving the policy after just *one* evaluation sweep instead of waiting for full convergence. Is this still valid? What is this variant called?

Q6. Monte Carlo prediction requires complete episodes to compute returns. Describe one type of task where this is a limitation and briefly explain why. How might you work around it?

Q7. An agent produces the episode: $S_0 = A, R_1 = 3; S_1 = B, R_2 = -1; S_2 = A, R_3 = 2$; terminal. Use $\gamma = 1$. (a) Compute G_2, G_1, G_0 . (b) Under first-visit Monte Carlo, which return is used to update $V(A)$?

Q8. Monte Carlo control estimates $Q(s, a)$ by averaging returns over episodes. Why must every (s, a) pair be visited sufficiently often for MC control to converge? Name one strategy that guarantees this.

Q9. State S has been visited 4 times with current estimate $V(S) = 6$. In a new episode, the observed return is $G = 10$. (a) Compute the new $V(S)$ using the sample-average update rule. (b) Compute the new $V(S)$ using a constant step size $\alpha = 0.1$. (c) Which update reacts more strongly to the new return, and when might a constant step size be preferred?

Q10. Over 3 episodes, the returns observed for state S and action $a = \text{left}$ are $G = 2, G = 8, G = 5$. (a) Compute the sample-average estimate $Q(S, \text{left})$ after all 3 episodes. (b) If $Q(S, \text{right}) = 4$, which action does a greedy policy select?